John Wheeler and his coworkers were among the earliest researchers to investigate wormholes, and they discovered many of their fundamental mathematical properties. More recently, though, Thorne and his collaborators revealed the full richness of wormholes by realizing that not only can they provide shortcuts through space, they can also provide shortcuts through time.

Here's the idea. Imagine that Bart and Lisa are standing at opposite ends of Springfield's wormhole—Bart at the power plant, Lisa at the Kwik-E-Mart—idly chatting with each other about what to get Homer for his birthday, when Bart decides to take a short transgalactic jaunt (to get Homer some of his favorite Andromedean fish fingers). Lisa doesn't feel up for the ride but, as she's always wanted to see Andromeda, she persuades Bart to load his wormhole mouth on his ship and take it along, so she can have a look. You might expect this to mean that Bart will have to keep stretching the wormhole longer as his journey progresses, but that assumes the wormhole connects the Kwik-E-Mart and Bart's ship through ordinary space. It doesn't. And, as illustrated in Figure 15.4, through the wonders of general relativistic geometry, the wormhole's length can remain fixed throughout the entire voyage. This is a key point. Even though Bart rockets off to Andromeda, his distance to Lisa through the wormhole does not change. This makes manifest the wormhole's role as a shortcut through space.

For definiteness, let's say that Bart heads off at 99.999999999999999 percent of light speed and travels four hours outbound to Andromeda, all the while continuing to chat with Lisa through the wormhole, just as they'd been doing before the flight. When